Modeling Data

Dominguez Center for Data Science

2026-05-29

Questions Round-Up

As we wrap up...

- I'd recommend downloading `R` and RStudio to your personal computer. You can find directions here.
- Make sure to export all your materials from Posit Cloud in the next week or so. I will delete the materials on June 30th.
- Attend the weekly **Writing/Coding Sessions**: 2:00 - 4:00pm on Mondays in June & July, Taylor Hall 203.

Models/Tests we are going to cover:

- For modeling a quantitative variable:
  - When you have a single categorical explanatory variable (with two categories): Two sample t-test
  - When you have 1 or more explanatory variables (categorical and/or quantitative): Linear regression
- For modeling a categorical variable:
  - When you have a single categorical explanatory variable: Chi-squared test
  - When you have 1 or more explanatory variables (categorical and/or quantitative): Logistic regression and classification trees

For the sake of time today, not going to focus on model assumptions, but make sure to check them!

Load Packages

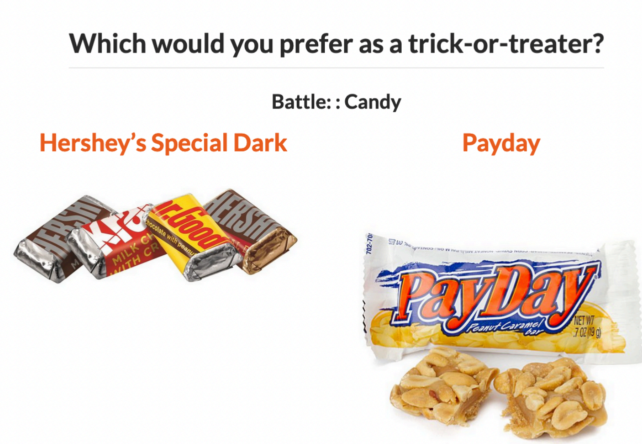
```
# For data wrangling and plotting
library(tidyverse)

# For fitting classification trees
library(rpart)

# For creating charts of trees
library(rpart.plot)
```

Data Example:

> "The social contract of Halloween is simple: Provide adequate treats to costumed masses, or be prepared for late-night tricks from those dissatisfied with your offer. To help you avoid that type of vengeance, and to help you make good decisions at the supermarket this weekend, we wanted to figure out what Halloween candy people most prefer. So we devised an experiment: Pit dozens of fun-sized candy varietals against one another, and let the wisdom of the crowd decide which one was best." – Walt Hickey
>
> "While we don't know who exactly voted, we do know this: 8,371 different IP addresses voted on about 269,000 randomly generated matchups.2 So, not a scientific survey or anything, but a good sample of what candy people like."



```r
candy <- read_csv("https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-power-
  mutate(pricepercent = pricepercent*100)


glimpse(candy)
```

```
Rows: 85
Columns: 13
$ competitorname   <chr> "100 Grand", "3 Musketeers", "One dime", "One quarter~
$ chocolate        <dbl> 1, 1, 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,~
$ fruity           <dbl> 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1,~
$ caramel          <dbl> 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0,~
$ peanutyalmondy   <dbl> 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
$ nougat           <dbl> 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,~
$ crispedricewafer <dbl> 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
$ hard             <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1,~
$ bar              <dbl> 1, 1, 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,~
$ pluribus         <dbl> 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 1, 1, 0, 1, 0, 1,~
$ sugarpercent     <dbl> 0.732, 0.604, 0.011, 0.011, 0.906, 0.465, 0.604, 0.31~
$ pricepercent     <dbl> 86.0, 51.1, 11.6, 51.1, 51.1, 76.7, 76.7, 51.1, 32.5,~
$ winpercent       <dbl> 66.97173, 67.60294, 32.26109, 46.11650, 52.34146, 50.~
```

T-test

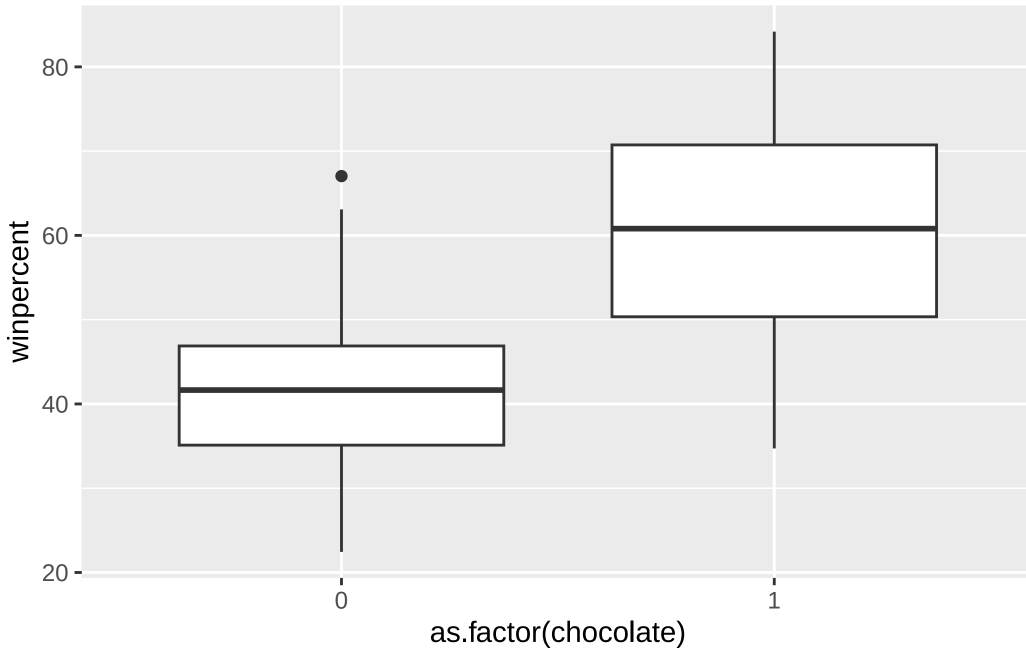
Consider this test when:

- Response variable (y): quantitative
- Explanatory variable (x): categorical with two categories

Let's say we want to predict the win percentage. Which of the categorical variables do you think would be a good predictor of the average win percentage?

First graph the data.

```r
ggplot(data = candy, mapping = aes(x = as.factor(chocolate), y = winpercent)) +
  geom_boxplot()
```



Then run the two-sample t-test.

```
t.test(winpercent ~ chocolate, data = candy)
```

```

	Welch Two Sample t-test

data:  winpercent by chocolate
t = -7.3031, df = 67.539, p-value = 4.164e-10
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -23.91110 -13.64744
sample estimates:
mean in group 0 mean in group 1 
       42.14226        60.92153 
```

Is the variable we selected a good predictor of the average price?

Multiple Linear Regression

Consider this model when:

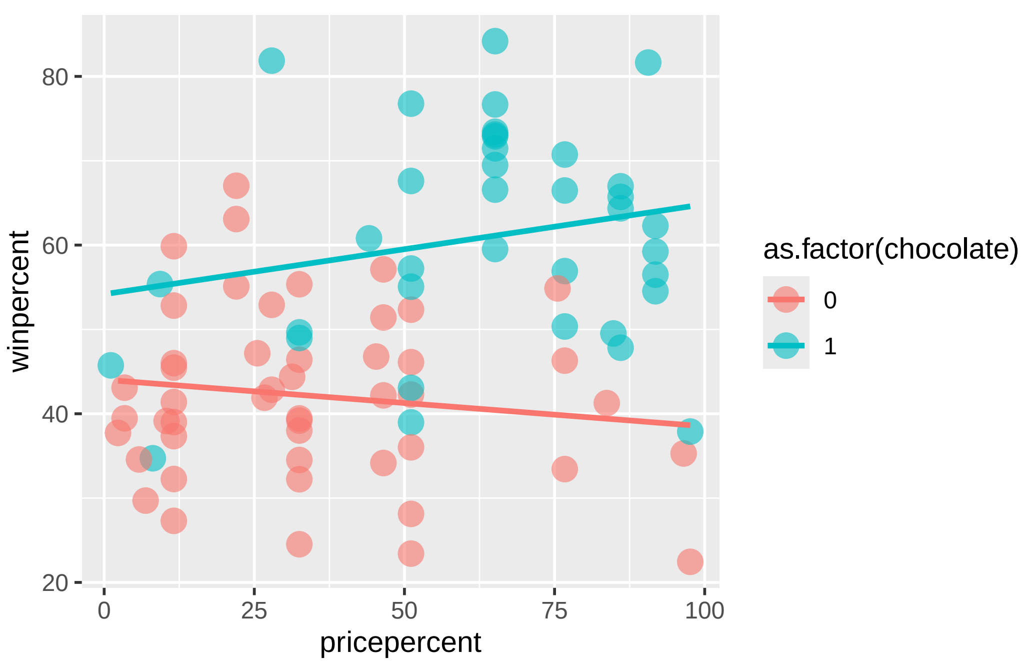
- Response variable (y): quantitative
- Explanatory variable(s) (x)'s: quantitative and/or categorical (simple = 1 explanatory variable)

And the relationship between the x's and y is reasonably approximated by the regression equation:

$$y = \beta_o + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

Let's now build a model for win percentage using price percentage and whether or not the candy contains chocolate. But first, we should graph the data.

```
ggplot(data = candy,
       mapping = aes(x = pricepercent,
                     y = winpercent, color = as.factor(chocolate))) +
  geom_point(alpha = 0.6, size = 4) +
  geom_smooth(method = "lm", se = FALSE)
```



Equal slopes model

You need to first determine if you want to have the same slope for candies with chocolate as for candies without. If you want the same slope, that is called the equal slopes model.

```r
mod_equal_slopes <- lm(winpercent ~ pricepercent + as.factor(chocolate),
                       data = candy)
summary(mod_equal_slopes)
```

```

Call:
lm(formula = winpercent ~ pricepercent + as.factor(chocolate),
    data = candy)

Residuals:
     Min       1Q   Median       3Q      Max
-25.2821  -7.5946  -0.1665   8.5309  25.1001

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)           41.57128    2.40375  17.294  < 2e-16 ***
pricepercent           0.01665    0.05077   0.328    0.744
as.factor(chocolate)1 18.29798    2.90880   6.291 1.46e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.48 on 82 degrees of freedom
Multiple R-squared:  0.4059,	Adjusted R-squared:  0.3914
F-statistic: 28.02 on 2 and 82 DF,  p-value: 5.337e-10
```

Different slopes model

If you want to allow the slopes to vary, that is called a different slopes model.

```r
mod_diff_slopes <- lm(winpercent ~ pricepercent * as.factor(chocolate),
                       data = candy)
summary(mod_diff_slopes)
```

```

Call:
lm(formula = winpercent ~ pricepercent * as.factor(chocolate),
    data = candy)

Residuals:
    Min      1Q  Median      3Q     Max
```

```
-26.705  -7.482  -0.658   8.361  24.715

Coefficients:
                                   Estimate Std. Error t value Pr(>|t|)    
(Intercept)                        44.03751    2.83637  15.526   <2e-16 ***
pricepercent                       -0.05525    0.06744  -0.819   0.4150    
as.factor(chocolate)1              10.13569    5.85807   1.730   0.0874 .  
pricepercent:as.factor(chocolate)1  0.16200    0.10123   1.600   0.1134    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.37 on 81 degrees of freedom
Multiple R-squared:  0.4241,	Adjusted R-squared:  0.4028 
F-statistic: 19.89 on 3 and 81 DF,  p-value: 9.413e-10
```

Prediction

```r
twix <- filter(candy, competitorname == "Twix")

predict(mod_diff_slopes, newdata = twix)
```

```
       1 
63.84475 
```

Logistic Regression

Consider this model when:

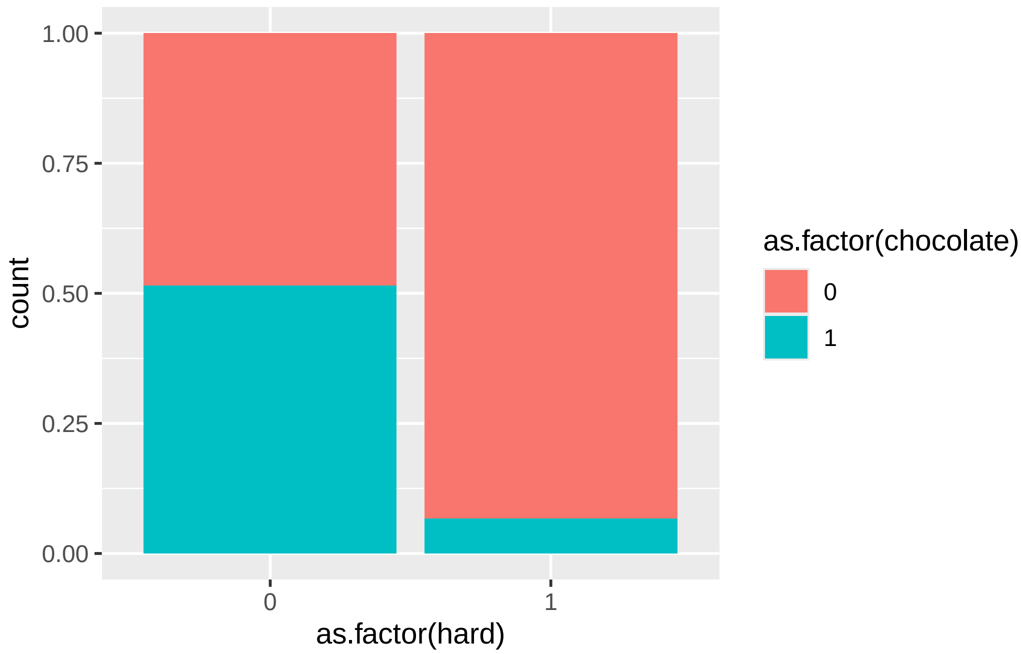
- Response variable (y): categorical (binary)
- Explanatory variable (x): quantitative and/or categorical

Question of interest: Does this candy have chocolate in it??

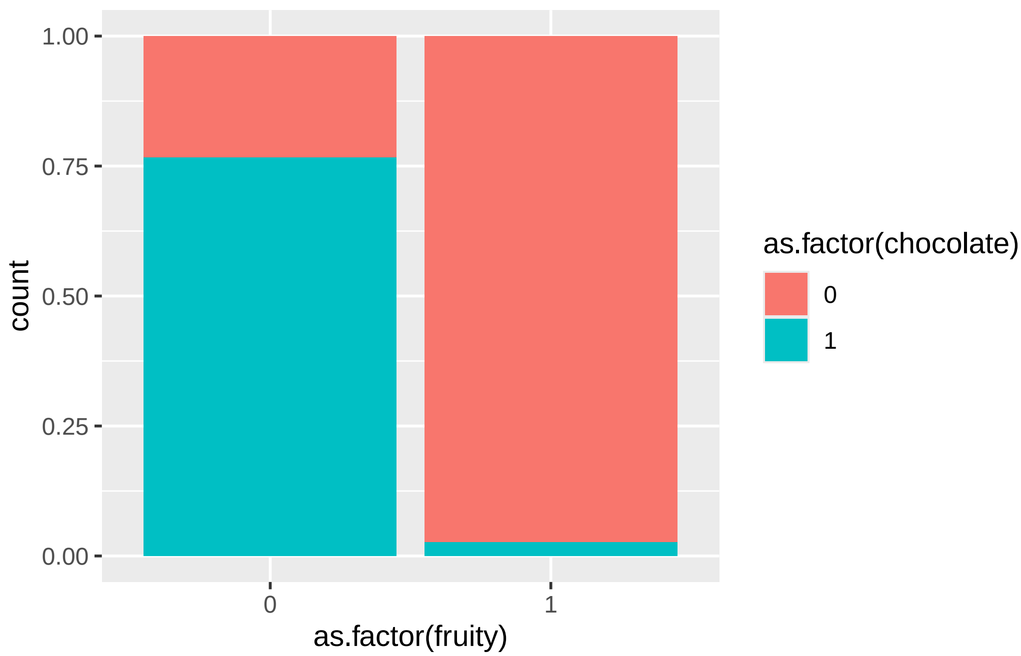
Follow-up question: Which variables in our dataset do you think would be good predictors of whether or not a candy will have chocolate in it?

Visualize the relationships between the variables

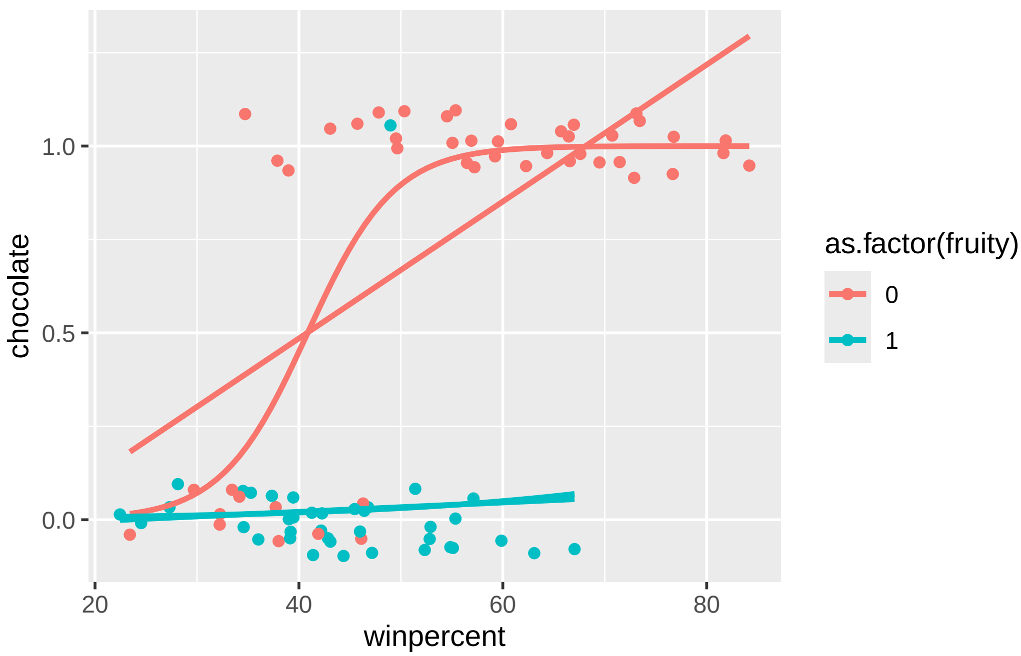
```
# Potential graphs
ggplot(data = candy, mapping = aes(x = as.factor(hard),
                                   fill = as.factor(chocolate))) +
  geom_bar(position = "fill")
```




```
ggplot(data = candy, mapping = aes(x = as.factor(fruity),
                                   fill = as.factor(chocolate))) +
  geom_bar(position = "fill")
```




```
ggplot(data = candy, mapping = aes(x = winpercent,
                                   y = chocolate,
                                   color = as.factor(fruity))) +
  geom_jitter(width = 0, height = 0.1) +
  geom_smooth(method = "lm", se = FALSE) +
  geom_smooth(method = "glm", method.args = list(family = "binomial"),
              se = FALSE)
```



Build a logistic regression model

```
# Linear model
mod_lin <- lm(chocolate ~ as.factor(fruity) + winpercent, data = candy)
summary(mod_lin)
```

```
Call:
lm(formula = chocolate ~ as.factor(fruity) + winpercent, data = candy)

Residuals:
     Min       1Q   Median       3Q      Max
-0.63922 -0.14170  0.01374  0.17507  0.90544

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)        -0.010511   0.129843  -0.081    0.936
as.factor(fruity)1 -0.582356   0.065592  -8.879 1.26e-13 ***
winpercent          0.014034   0.002229   6.295 1.44e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.278 on 82 degrees of freedom
Multiple R-squared:  0.6967,    Adjusted R-squared:  0.6893
F-statistic: 94.18 on 2 and 82 DF,  p-value: < 2.2e-16
```

```r
# Logistic model
mod_log <- glm(chocolate ~ as.factor(fruity) + winpercent, data = candy,
               family = "binomial")
summary(mod_log)
```

```

Call:
glm(formula = chocolate ~ as.factor(fruity) + winpercent, family = "binomial",
    data = candy)

Coefficients:
                   Estimate Std. Error z value Pr(>|z|)
(Intercept)        -7.62478    2.40682  -3.168 0.001535 **
as.factor(fruity)1 -5.79549    1.48680  -3.898  9.7e-05 ***
winpercent          0.18799    0.05552   3.386 0.000709 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 116.407  on 84  degrees of freedom
Residual deviance:  32.104  on 82  degrees of freedom
AIC: 38.104

Number of Fisher Scoring iterations: 7
```

Prediction

You can use the `predict()` function in the same way as we did for linear regression.

```r
new_cases <- data.frame(fruity = 0, winpercent = 50)
predict(mod_log, newdata = new_cases, type = "response")
```

```
        1
0.8550401
```

Classification Trees

Consider this model when:

- Response variable (y): categorical
- Explanatory variable (x): quantitative and/or categorical

This is a more flexible model fit than logistic regression.

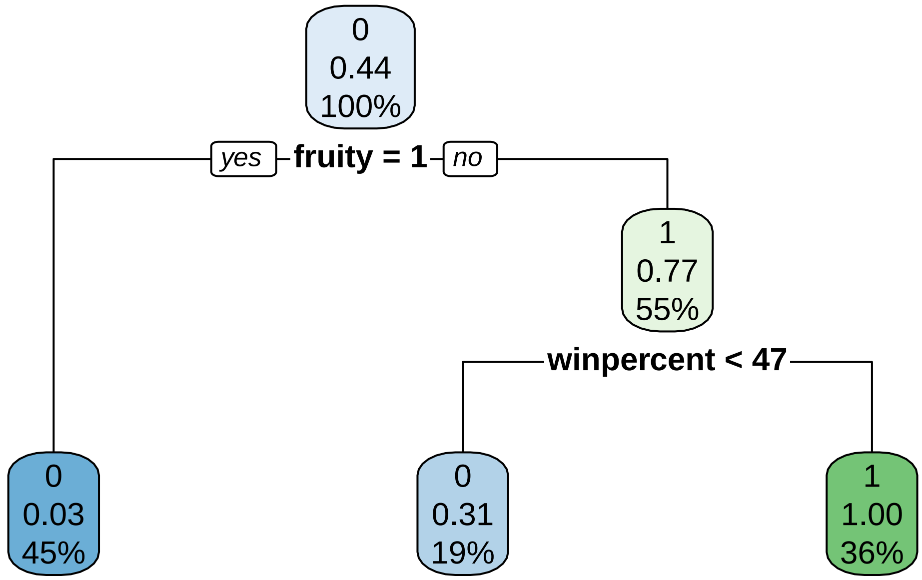
```r
# Fit the model
tree <- rpart(chocolate ~ . - competitorname, data = candy, method = "class")
tree
```

```
n= 85

node), split, n, loss, yval, (yprob)
      * denotes terminal node

1) root 85 37 0 (0.56470588 0.43529412)
  2) fruity>=0.5 38  1 0 (0.97368421 0.02631579) *
  3) fruity< 0.5 47 11 1 (0.23404255 0.76595745)
    6) winpercent< 47.06318 16  5 0 (0.68750000 0.31250000) *
    7) winpercent>=47.06318 31  0 1 (0.00000000 1.00000000) *
```

```r
rpart.plot(tree)
```



Which model is better: the logistic regression model or the classification tree?

Three metrics:

- Accuracy: Proportion of the guesses were correct
- Sensitivity: Among the chocolate candies, proportion of times that the model identified the candy has chocolate
- Specificity: Among the non-chocolate candies, proportion of times that the model identified the candy doesn't have chocolate

```
# How do we access the predictions?
mod_log$fitted.values
```

```
           1            2            3            4            5            6 
0.9930717593 0.9938421950 0.1736410990 0.7397432702 0.0271030505 0.8629517717 
           7            8            9           10           11           12 
0.9558307979 0.0383285038 0.3824594355 0.0009757615 0.4260742583 0.0012931995 
          13           14           15           16           17           18 
0.0001492352 0.0041786848 0.0024674304 0.0048688710 0.0023433814 0.0097036752 
          19           20           21           22           23           24 
0.0640219308 0.2308960271 0.0228591193 0.0041063273 0.9418718857 0.9834396396 
          25           26           27           28           29           30 
0.9523403317 0.9709987112 0.0002937128 0.9581869379 0.9988951129 0.0035421407 
```

```
          31           32           33           34           35           36 
0.0023239271 0.0300759597 0.9970112146 0.9925387186 0.0090548161 0.9385829224 
          37           38           39           40           41           42 
0.9978001523 0.9782278139 0.9887154063 0.7968476718 0.9324921076 0.0467840884 
          43           44           45           46           47           48 
0.9965734663 0.9923926640 0.0001009495 0.0024610736 0.7462081297 0.9956682326 
          49           50           51           52           53           54 
0.3697286677 0.0034607785 0.0016602131 0.9995759405 0.9997254840 0.9979343096 
          55           56           57           58           59           60 
0.9977110625 0.0011282190 0.9912437059 0.1149850852 0.0046552731 0.2502249957 
          61           62           63           64           65           66 
0.1735126952 0.0447294856 0.3769790426 0.0083795584 0.9988752643 0.9725108551 
          67           68           69           70           71           72 
0.1028017327 0.0296109068 0.3062080104 0.0009870617 0.2076949961 0.1728309046 
          73           74           75           76           77           78 
0.0002515951 0.0428208726 0.0145996040 0.6157899065 0.7257656829 0.8467783120 
          79           80           81           82           83           84 
0.0104339048 0.9995577649 0.0075915868 0.0022683073 0.0061928604 0.5628620667 
          85 
0.8435957161 
```

```
predict(tree, type = "class")
```

```
 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 
 1  1  0  0  0  1  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  1  1  1 
27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 
 0  1  1  0  0  0  1  1  0  1  1  1  1  1  1  0  1  1  0  0  0  1  0  0  0  1 
53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 
 1  1  1  0  1  0  0  0  0  0  0  0  1  1  0  0  0  0  0  0  0  0  0  0  0  1 
79 80 81 82 83 84 85 
 0  1  0  0  0  0  1 
Levels: 0 1
```

```
# Add predictions
candy <- candy %>%
  mutate(log_model_guess = if_else(mod_log$fitted.values >= 0.5, 1, 0),
         tree_model_guess = predict(tree, type = "class"))

# Confusion matrix
count(candy, chocolate, log_model_guess)
```

```
# A tibble: 4 x 3
```

```
  chocolate log_model_guess     n
      <dbl>           <dbl> <int>
1         0               0    45
2         0               1     3
3         1               0     4
4         1               1    33
```

```
count(candy, chocolate, tree_model_guess)
```

```
# A tibble: 3 x 3
  chocolate tree_model_guess     n
      <dbl> <fct>            <int>
1         0 0                   48
2         1 0                    6
3         1 1                   31
```

Resources related to modeling

We have just scratched the surface of building and interpreting models in `R`. Here are some additional resources on modeling:

- Chapters 5, 6, and 10 of ModernDive
- For a more machine learning approach, check out Tidy Modeling with R or Introduction to Statistical Learning in R
- For more advanced modeling, check out Beyond Multiple Linear Regression
- For bayesian model, check out Bayes Rules!

Homework

- Download materials from Posit Cloud.
- Go back over all the materials we have covered and try to apply to your own work.
- Bring ideas to the weekly **Writing/Coding Sessions**: 2:00 - 4:00pm on Mondays in June & July, Taylor Hall 203.